# STAT6611
# Lecture 02 (R, Aug 27)

- No homework yet
- Office hours soon

Later: normal equations $X'Xb = X'y$ $\quad X'XB = X'$ more general

**Definition.** The equations $Ax = y$ are said to be consistent if they have at least one solution.

$A$: $m \times n$, $x$: $n \times 1$, $y$: $m \times 1$

ie. consistent means (given $A$ & $y$) that $y \in \mathbb{R}^m$ is a linear combination of the $n$ columns of $A$.

**Theorem 1.2.** *For any matrix $A$, a matrix $G$ is a generalized inverse of $A$ if and only if $x = Gy$ is a solution to the equations $Ax = y$ for every vector $y$ for which the equations are consistent.*

ie. $G = A^-$ $\Leftrightarrow$ $x = Gy$ is a solution to $Ax = y$ provided any solution exists (ie. $A(Gy) = y$)

$\Leftarrow$ If $m \times n$ $A = [a_1, \ldots, a_n]$, each column $a_i \in \mathbb{R}^m$ gives a consistent equation: $Ax_i = a_i$ holds where $x_i$ is $i$th coordinate vector (1 in $i$th component)

So, by assumption, it must be that $Ga_i$ must be a solution to $Ax = a_i$ too, for each $i = 1, \ldots n$.

That is, $A(Ga_i) = a_i$ for each $i = 1, \ldots, n$.

Hence, $AGA = AG[a_1 \ldots a_n] = [a_1 \ldots a_n] = A$

So, by def, $G$ is g.i of $A$ ($G = A^-$)

$\Rightarrow$ Assume $G$ is g.i. of $A$ & suppose $Ax = y$ holds for some $x$.

Need show $Gy$ is a solution too.

Note $AGy = AG(Ax) = AGAx = Ax = y$

($Ax = y$; $G$ is g.i.)

generalize Theorem 1.2 to matrices Y (extending vector $\underline{y}$)

**Remark.** Let $A$ be an $m \times n$ matrix and $X$ an $n \times p$ matrix. Then, $Y = AX$ is an $m \times p$ matrix. Let $\underline{x}_k$ be the $k$th column of $X$ and $\underline{y}_k$ be the $k$th column of $Y$. Then,

$n \times p$ $m \times p$ $m \times n$

$$AX = Y \iff A(\underline{x}_1, \ldots, \underline{x}_p) = (\underline{y}_1, \ldots, \underline{y}_p) \iff A\underline{x}_k = \underline{y}_k \text{ for all } k.$$

Thus, Theorem 1.2 can be extended to the matrix equation $AX = Y$.

That is, for any matrix $A$, a matrix $G$ is a generalized inverse of $A$ if and only if $X = GY$ is a solution to the matrix equation $AX = Y$ for every matrix $Y$ for which the equation is consistent.

ie. $GY = G(\underline{y}_1 \ldots \underline{y}_p) = (G\underline{y}_1, \ldots, G\underline{y}_p)$ is a solution to $AX = Y$ provided a solution exists

**Theorem 1.3.** *All solutions to a consistent matrix equation $AX = Y$ are generated by putting*

$$X = A^-Y + (I - A^-A)Z$$

this is one solution based on a g.i of A

this part gives all other solutions

*and letting $Z$ range over the space of all possible matrices (of appropriate dimensions), where $I$ denotes the identity matrix (of appropriate dimensions).*

*Proof.* Based on the remark after Theorem 1.2, we have

(i.e. if $AX = Y$ has solution then $A^-Y$ must be a solution ie $A(A^-Y) = Y$)

plug-in form

$$A[A^-Y + (I - A^-A)Z] = A(A^-Y) + A(I - A^-A)Z = A(A^-Y) + (A - AA^-A)Z = A(A^-Y) = Y,$$

$= A$ by def of g.i.

theorem 1.2

for any $Z$ (of appropriate dimensions).

Conversely, for any solution $X^*$ to $AX = Y$, (i.e. $AX^* = Y$)

$$X^* = A^-Y + X^* - A^-Y$$
$$= A^-Y + X^* - A^-(AX^*)$$
$$= A^-Y + [I - A^-A]X^* \quad \text{form holds (ie } Z = X^*\text{)}$$

□

**Lemma 1.1.** *A matrix $G$ is a generalized inverse of a matrix $A$ if and only if $G'$ is a generalized inverse of $A'$, where $A'$ is the transpose of $A$.*

pf: $AGA = A \Leftrightarrow A' = (AGA)' = A'G'A'$

$G$ g.i of $A$ — $G'$ g.i of $A'$

**Corollary.** If $A$ is a symmetric matrix, then $(A^-)'$ is a generalized inverse of $A$. In particular, for any matrix $X$, $[(X'X)^-]'$ is a generalized inverse of $X'X$.

pf: $(A^-)'$ is g.i of $A' = A$ ↑ If $A$ is symmetric

If $(X'X)^-$ is g.i of $X'X$ then $((X'X)^-)'$ is g.i. too

**Remark.** A generalized inverse of a symmetric matrix $A$ can be nonsymmetric.

e.g. $A = O_{m\times m}$ is symmetric but $A^-$ could be any $m\times m$ matrix

If $A$ is symmetric, we could start from a g.i. $A^-$ & make a new & symmetric g.i $(A^- + (A^-)')/2$

**Lemma 1.2.** *For any matrix $X$, $\mathrm{tr}(X'X) = 0$ implies $X = 0$.*

↑ "trace"

*Proof.* Let $\underline{x}_j$ be the $j$th column of $X$. Note that

$$X'X = (\underline{x}_1, \ldots, \underline{x}_n)'(\underline{x}_1, \ldots, \underline{x}_n) = \begin{pmatrix} \underline{x}_1' \\ \vdots \\ \underline{x}_n' \end{pmatrix} (\underline{x}_1, \ldots, \underline{x}_n) = \begin{bmatrix} \underline{x}_1'\underline{x}_1 & \cdots & \underline{x}_1'\underline{x}_n \\ \vdots & & \vdots \\ \underline{x}_n'\underline{x}_1 & \cdots & \underline{x}_n'\underline{x}_n \end{bmatrix}.$$

$X'$ $X$

& $\mathrm{tr}(X'X) = \sum_{i=1}^n \underset{\sim}{x}_i' \underset{\sim}{x}_i = \sum_{i=1}^n \|\underset{\sim}{x}_i\|^2$

So $\mathrm{tr}(X'X) = 0 \Rightarrow$ each $\|\underset{\sim}{x}_i\|^2 = 0$ $i = 1..n$

$\Rightarrow$ each $\underline{x}_i = 0$ $i = 1..n \Rightarrow X = 0$

**Lemma 1.3.** *For any matrix $X$ and for any matrices $P$ and $Q$ of appropriate dimensions, we have*

→ *(i) $PX'X = QX'X$ implies $PX' = QX'$ and*

*(ii) $X'XP = X'XQ$ implies $XP = XQ$.*

("scratch-off" result)

$XXA = XXB \Rightarrow XA = XB$

$AX'X = BX'X \Rightarrow AX' = BX'$

*Proof of* (i): $PX'X = QX'X$ implies that

$$0 = (PX'X - QX'X)(P-Q)' = (XP'-XQ')'X(P'-Q') = (XP'-XQ')'(XP'-XQ').$$

suppose

difference

$= \tilde{X}'$ $\tilde{X}$ say

$\Rightarrow tr(\tilde{X}'\tilde{X}) = 0$

$\Rightarrow \tilde{X} = 0$ by last result

$\Rightarrow XP' = XQ' \Leftrightarrow PX' = QX'$

g.i of $X'$

**Lemma 1.4.** *For any matrix* $X$, $X'X(X'X)^{-}X' = X'$, *that is,* $X(X'X)^{-}$ *is a generalized inverse of* $X'$.

proof: $(X'X)(X'X)^{-}(X'X) = X'X$ by def

$\Rightarrow (X'X)(X'X)^{-}X' = X'$ Lemma 1.3

$\Rightarrow X(X'X)^{-}$ is g.i of $X'$ by def

**Corollary 1.** (i) $X'X[(X'X)^{-}]'X' = X'$, (ii) $X(X'X)^{-}X'X = X$, and (iii) $X[(X'X)^{-}]'X'X = X$.

**Corollary 2.** Corresponding to any matrix $X$, there exists a matrix $B$ satisfying $X'XB = X'$.